HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Owned module performance metrics across safety, quality, and output for a critical wafer packaging line, contributing to a **12% reduction** in yield loss over six months.
- Developed and deployed **Python scripts** to automate data collection and analysis for process capability studies, reducing manual data entry time by approximately **6 hours per week**.
- Collaborated with 24/7 technician teams to rapidly diagnose and contain P1/P2 incidents in a highly automated fab environment, ensuring minimal impact on production schedules.
- Partnered with global Virtual Factory teams to align manufacturing standards and deploy best practices for **process control**, improving consistency across multiple sites.
- Led structured problem-solving sessions using **statistical methods** and data analysis to identify root causes for process variation, resulting in a **15% decrease** in defect rates.
- Supported technology ramps by coordinating tool installs and validation activities, ensuring smooth transitions into high-volume manufacturing for new product introductions.
- Containerized and deployed **Python data analysis tools** to an **AWS Lambda** environment, enabling on-demand processing of manufacturing data streams.

SKILLS

Manufacturing & Process – Process Optimization, Process Control, Continuous Improvement, Statistical Methods, Problem Solving, Technology Ramps, Equipment Metrics, Yield Loss Reduction

Programming & Scripting – Python, SQL, Shell Scripting

Cloud & DevOps – AWS Lambda, Docker, CI/CD, Git

Data Analysis & Tools – JMP, Data Analysis, Data Mining, Data Warehousing, Power BI, Tableau, Pandas, NumPy

Software Engineering – Java, Spring Boot, REST APIs, PostgreSQL, MySQL, Oracle, Microservices

Methodologies & Operations – Agile, Scrum, SLA Management, Incident Response, Change Management, Production Support

PROJECTS

Automated Data Pipeline for Semiconductor Process Monitoring

- Engineered a **Python data pipeline** on **AWS Lambda** to ingest and process real-time sensor data from manufacturing equipment, handling approximately **5,000 events per hour**.
- Developed **Power BI dashboards** to visualize key performance indicators and process trends, enabling operations teams to identify anomalies and potential issues proactively.
- Implemented **SQL queries** against a **PostgreSQL** database to extract and transform historical process data for long-term trend analysis and reporting.

Statistical Analysis and Process Improvement for Enterprise Systems

- Applied **statistical methods** and **JMP software** to analyze historical operational data from enterprise resource planning systems, identifying key drivers of system downtime.
- Reduced process variation by **18%** through data-driven recommendations and implementation of targeted configuration changes, improving system stability.
- Collaborated with business analysts to translate technical findings into actionable insights for process optimization initiatives.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)